

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	28 June 2026
Team ID	Not Given
Project Name	Rising Waters
Maximum Marks	5 Marks

Team Members	Skill Wallet Team ID's	Roll Numbers (College: ACET)
Mohana Deepika Kancherla (Team Leader)	SWUID2026226898	23MH1A0596
Flora Disowja Sarugolu	SWUID2026212008	24P35A0549
Santhi Illa	SWUID2026212021	24P35A0550
Mathan Bhavesh Hari Raghava Krishna Munaga	SWUID2026212007	24P35A0532

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Product Backlog

The product backlog contains prioritized features, user stories, bug fixes, and enhancements required to develop the Rising Waters flood prediction system.

Sprint Schedule

The sprint schedule organizes project tasks into short development cycles, ensuring timely implementation, testing, review, and continuous project improvement.

Estimation

Project estimation evaluates development effort, resources, story points, and timelines to ensure efficient planning, execution, and successful project delivery.

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Authentication	USN-1	As a user, I can register and log in securely to access the flood prediction dashboard.	3	High	M, Deepika
Sprint-1	Weather Data Integration	USN-2	As a system, I collect real-time weather and rainfall data from APIs.	5	High	M, Bhavesh
Sprint-1	River Level Monitoring	USN-3	As a system, I collect river water level data from sensors/APIs.	5	High	Santhi Illa
Sprint-2	AI Flood Prediction	USN-4	As a user, I receive flood risk predictions generated by the AI model.	8	High	Flora Disowja
Sprint-2	Risk Classification	USN-5	As a system, I classify flood risk into Low, Medium, and High levels.	5	High	M .Bhavesh
Sprint-2	Dashboard	USN-6	As a user, I can view flood risk maps, weather data, and prediction reports.	5	Medium	M, Deepika
Sprint-3	Alert System	USN-7	As a user, I receive SMS, email, and mobile notifications during flood emergencies.	8	High	Flora Disowja
Sprint-3	Report Generation	USN-8	As an authority, I can generate reports and analyze flood trends.	5	Medium	Santhi Illa
Sprint-4	System Testing	USN-9	As a developer, I validate prediction accuracy, performance, and reliability.	5	High	Entire Team
Sprint-4	Deployment	USN-10	As an administrator, I deploy the solution on the cloud for public use.	6	High	Entire Team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	18	7 Days	01 Aug 2025	07 Aug 2025	18	07 Aug 2025

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-2	18	7 Days	08 Aug 2025	14 Aug 2025	18	14 Aug 2025
Sprint-3	18	7 Days	15 Aug 2025	21 Aug 2025	18	21 Aug 2025
Sprint-4	16	7 Days	22 Aug 2025	28 Aug 2025	16	28 Aug 2025

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart: A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>